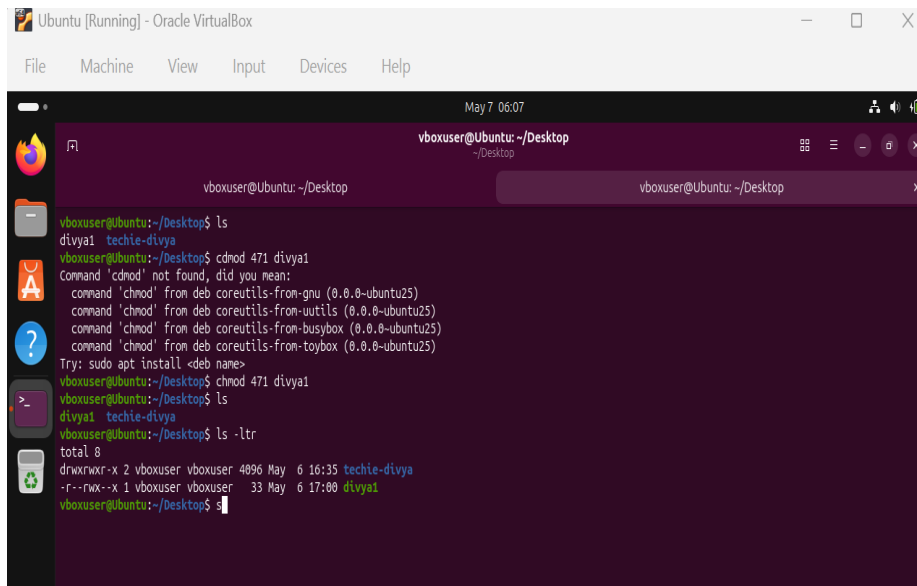


Linux Commands Day 2

Changed the permissions of the file “divya1” using the command “chmod”



```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

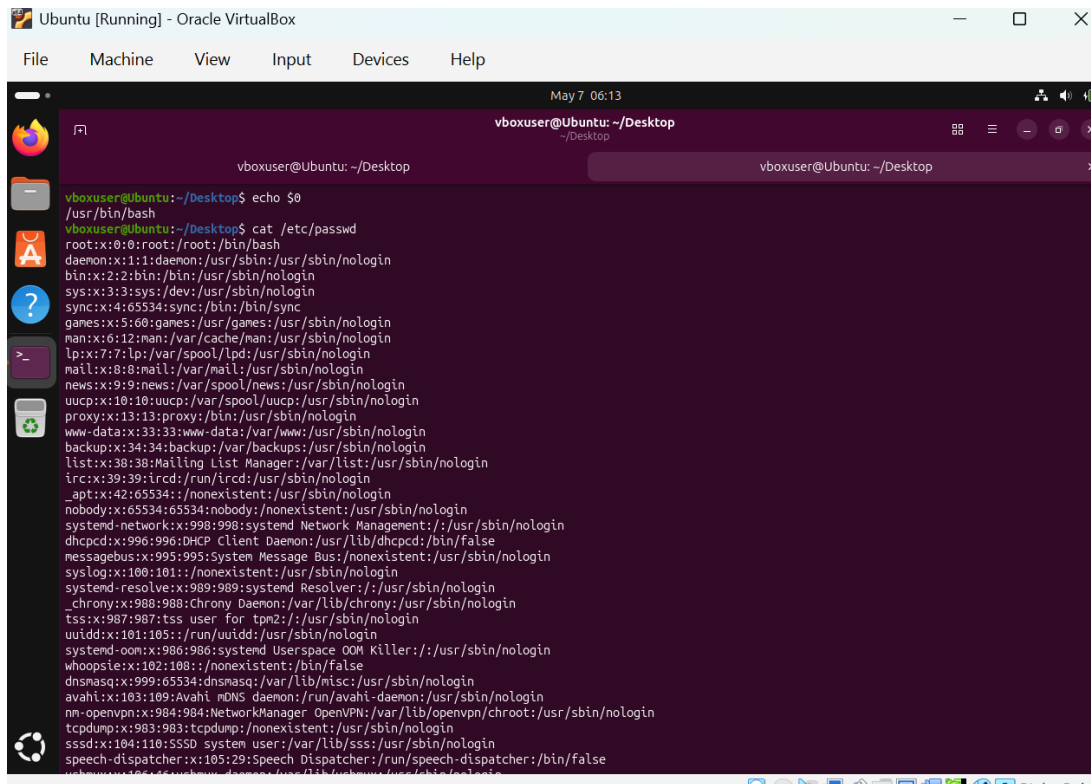
vboxuser@Ubuntu: ~/Desktop
vboxuser@Ubuntu: ~/Desktop

vboxuser@Ubuntu:~/Desktop$ ls
divya1 techie-divya
vboxuser@Ubuntu:~/Desktop$ chmod 471 divya1
Command 'chmod' not found, did you mean:
  command 'chmod' from deb coreutils-from-gnu (0.0.0-ubuntu25)
  command 'chmod' from deb coreutils-from-utils (0.0.0-ubuntu25)
  command 'chmod' from deb coreutils-from-busybox (0.0.0-ubuntu25)
  command 'chmod' from deb coreutils-from-toybox (0.0.0-ubuntu25)
Try: sudo apt install <deb name>
vboxuser@Ubuntu:~/Desktop$ chmod 471 divya1
vboxuser@Ubuntu:~/Desktop$ ls
divya1 techie-divya
vboxuser@Ubuntu:~/Desktop$ ls -ltr
total 8
drwxrwxr-x 2 vboxuser vboxuser 4096 May  6 16:35 techie-divya
-r--rwx--x 1 vboxuser vboxuser  33 May  6 17:00 divya1
vboxuser@Ubuntu:~/Desktop$
```

Echo \$0 shows the currently working shell or the script

Here it is /usr/bin/bash shell

To check the list of users “cat /etc/passwd “, there are two users here root and vboxuser in /bash shell



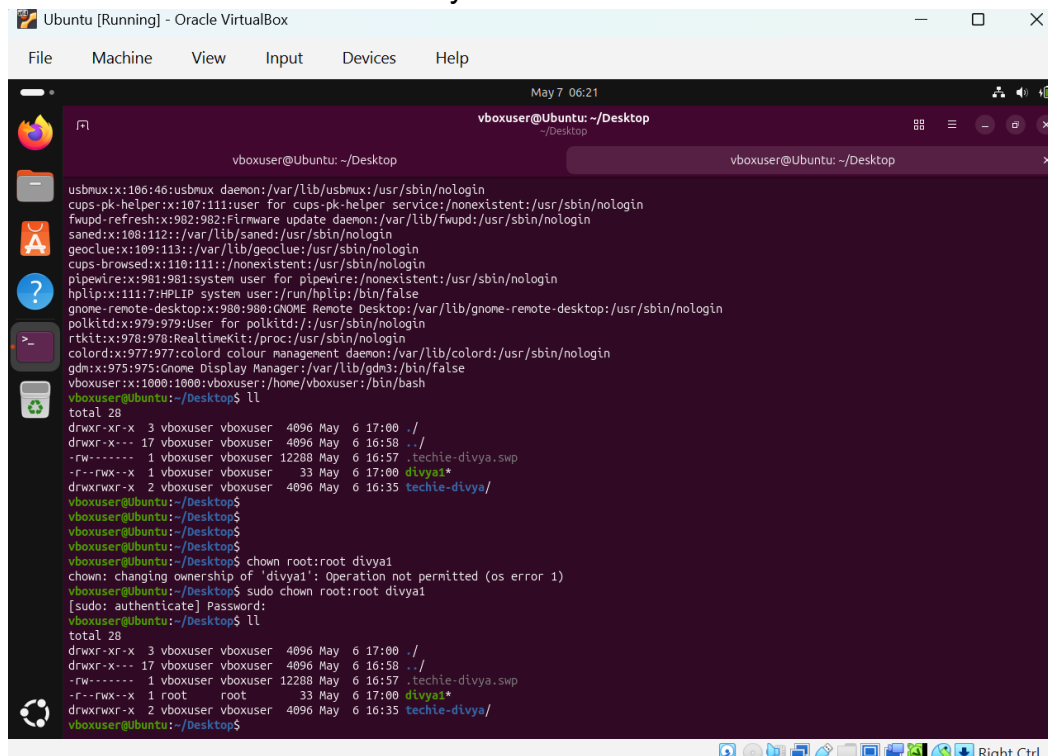
```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@Ubuntu: ~/Desktop
vboxuser@Ubuntu: ~/Desktop

vboxuser@Ubuntu:~/Desktop$ echo $0
/usr/bin/bash
vboxuser@Ubuntu:~/Desktop$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
chrony:x:988:988:chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:/:/usr/sbin/nologin
uidd:x:101:105::/run/uidd:/usr/sbin/nologin
systemd-oom:x:986:986:systemd Userspace OOM Killer:/:/usr/sbin/nologin
whoopsie:x:102:108::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:103:109:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
nm-openvpn:x:984:984:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
tcpdump:x:983:983:tcpdump:/nonexistent:/usr/sbin/nologin
sssd:x:104:110:SSSD system user:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:105:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
vboxuser:x:106:106:vboxuser:/home/vboxuser:/bin/bash
```

The command “Chown” changes the owner of a file. Here, I have changed the owner of the file divya1 from “vboxuser” to “root”.

Command used is `chown root:root divya1`



```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@Ubuntu: ~/Desktop
vboxuser@Ubuntu: ~/Desktop

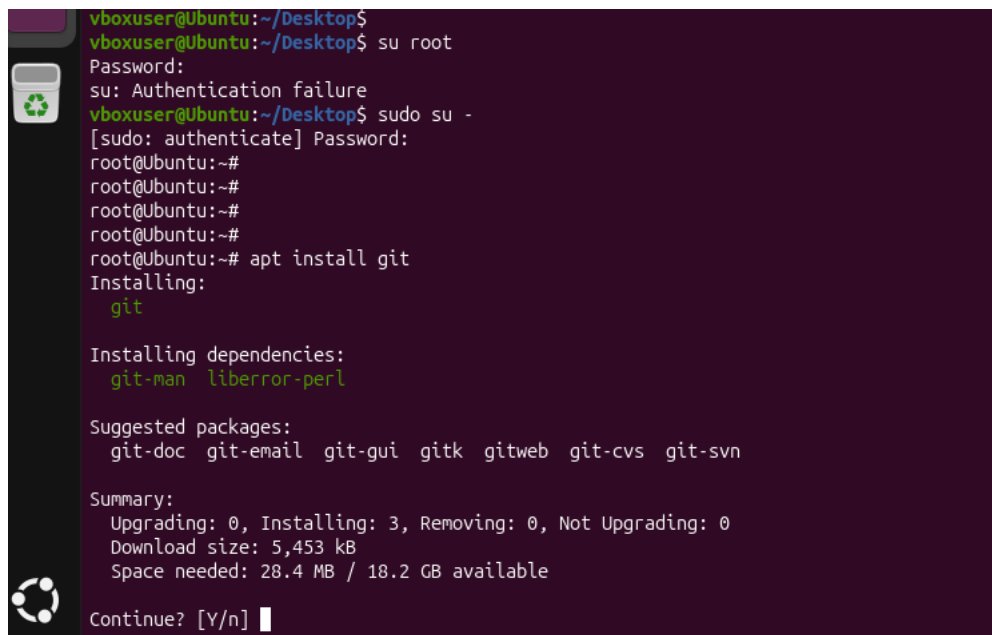
usbmux:x:106:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:107:111:user for cups-pk-helper service:/usr/sbin/nologin
fwupd-refresh:x:982:982:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:108:112:./var/lib/saned:/usr/sbin/nologin
geoclue:x:109:113:./var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:110:111:./nonexistent:/usr/sbin/nologin
pipewire:x:981:981:system user for pipewire:/nonexistent:/usr/sbin/nologin
hplip:x:111:7:HPLIP system user:/run/hplip/bin/false
gnome-remote-desktop:x:980:980:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:979:979:User for polkitd:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3/bin/false
vboxuser:x:1000:1000:vboxuser:/home/vboxuser/bin/bash

vboxuser@Ubuntu:~/Desktop$ ll
total 28
drwxr-xr-x 3 vboxuser vboxuser 4096 May 6 17:00 ./
drwxr-xr-x 17 vboxuser vboxuser 4096 May 6 16:58 ../
-rw-r--r-- 1 vboxuser vboxuser 12288 May 6 16:57 .techie-divya.swp
-r--rwx--x 1 vboxuser vboxuser 33 May 6 17:00 divya1*
drwxrwxr-x 2 vboxuser vboxuser 4096 May 6 16:35 techie-divya/

vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$ chown root:root divya1
chown: changing ownership of 'divya1': Operation not permitted (os error 1)
vboxuser@Ubuntu:~/Desktop$ sudo chown root:root divya1
[sudo: authenticate] Password:
vboxuser@Ubuntu:~/Desktop$ ll
total 28
drwxr-xr-x 3 vboxuser vboxuser 4096 May 6 17:00 ./
drwxr-xr-x 17 vboxuser vboxuser 4096 May 6 16:58 ../
-rw-r--r-- 1 vboxuser vboxuser 12288 May 6 16:57 .techie-divya.swp
-r--rwx--x 1 root root 33 May 6 17:00 divya1*
drwxrwxr-x 2 vboxuser vboxuser 4096 May 6 16:35 techie-divya/

vboxuser@Ubuntu:~/Desktop$
```

To switch the root user “`sudo su -`”



```
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$ su root
Password:
su: Authentication failure
vboxuser@Ubuntu:~/Desktop$ sudo su -
[sudo: authenticate] Password:
root@Ubuntu:~#
root@Ubuntu:~#
root@Ubuntu:~#
root@Ubuntu:~#
root@Ubuntu:~# apt install git
Installing:
  git

Installing dependencies:
  git-man liberror-perl

Suggested packages:
  git-doc git-email git-gui gitk gitweb git-cvs git-svn

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 0
  Download size: 5,453 kB
  Space needed: 28.4 MB / 18.2 GB available

Continue? [Y/n]
```

To add a user or create a new user, “useradd techie_divya”. User has been created.

```
vboxuser@Ubuntu:~/Desktop$ sudo su -
[sudo: authenticate] Password:
root@Ubuntu:~# useradd techie_divya
root@Ubuntu:~# cat /etc/passwd
cat: /etc/passwd: No such file or directory
root@Ubuntu:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
tss:x:987:987:tss user for tpm2:/:/usr/sbin/nologin
uuidd:x:101:105:/:run/uuidd:/usr/sbin/nologin
systemd-oom:x:986:986:systemd Userspace OOM Killer:/:/usr/sbin/nologin
avahi:x:103:109:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
nm-openvpn:x:984:984:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
tcpdump:x:983:983:tcpdump:/nonexistent:/usr/sbin/nologin
sssd:x:104:110:SSSD system user:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:105:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:106:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:107:111:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:982:982:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:108:112:/:var/lib/saned:/usr/sbin/nologin
geoclue:x:109:113:/:var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:110:111:/:nonexistent:/usr/sbin/nologin
pipewire:x:981:981:system user for pipewire:/nonexistent:/usr/sbin/nologin
hplip:x:111:7:HPLIP system user:/run/hplip:/bin/false
gnome-remote-desktop:x:980:980:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:979:979:User for polkitd:/:/usr/sbin/nologin
rtkit:x:978:978:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:977:977:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
vboxuser:x:1000:1000:vboxuser:/home/vboxuser:/bin/bash
techie_divya:x:1001:1001:/:home/techie_divya:/bin/sh
root@Ubuntu:~#
```

To create a password for the new user “passwd tech_divya”

```
root@Ubuntu:~# passwd techie_divya
New password:
Retype new password:
passwd: password updated successfully
root@Ubuntu:~#
```

To check the user which we are in “id” command is used.

```
root@Ubuntu:~# id
uid=0(root) gid=0(root) groups=0(root)
root@Ubuntu:~#
```

To find the path of the sudoers file, “find / -name sudoers”

```
find: '/snap/core24/1587/var/lib/snapd/void': Permission denied
vboxuser@Ubuntu:~/Desktop$ sudo su -
root@Ubuntu:~# find / -name sudoers
/etc/sudoers
/usr/share/doc/sudo/examples/sudoers
/usr/share/sudo-common/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
/snap/core24/1587/etc/sudoers
root@Ubuntu:~# vi /etc/sudoers
```

Cat /etc/sudoers command is used to check the content inside the /etc/sudoers file

```
root@Ubuntu:~# cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"
```

Now we need to edit the /etc/sudoers file and provide extra privileges to the techie_divya user. Command used is vi /etc/sudoers.

and insert the line techie_divya ALL=(ALL:ALL) ALL under user privilege Specifications

```
# User privilege specification
root    ALL=(ALL:ALL) ALL
techie_divya ALL=(ALL:ALL) ALL
# Members of the admin group may gain root privileges
```

“su” is the command used to switch user. Here, I have switched to techie_divya by using “su techie_divya” and tried to install git using the command “sudo apt install git”. It has worked successfully as the user got extra privileges.

```
root@Ubuntu:~#
root@Ubuntu:~#
root@Ubuntu:~# su techie_divya
$ apt install git
Error: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
$ sudo apt install git
[sudo: authenticate] Password:
Installing:
git

Installing dependencies:
git-man liberror-perl

Suggested packages:
git-doc git-email git-gui gitk gitweb git-cvs git-svn

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 0
Download size: 5,453 kB
Space needed: 28.4 MB / 18.2 GB available

Continue? [Y/n]
```

I/O Redirections:

Echo is used to display the exact content as output

Echo "Hello World" gives output as Hello World

Single Redirection (>) – I have run (echo "Hello World" > test) this command so the file got created and content got added to the file.

```
logout
vboxuser@Ubuntu:~/Desktop$ echo "Hello World"
Hello World
vboxuser@Ubuntu:~/Desktop$ echo "Hello World" > test
vboxuser@Ubuntu:~/Desktop$ ls
divya1  techie-divya  test
vboxuser@Ubuntu:~/Desktop$ cat test
Hello World
vboxuser@Ubuntu:~/Desktop$
```

So now when I do echo for the same test file with different content, the new content will override the old content . Suppose I add new content as "This is divya Erupaka" the old content "Hello World" gets overridden or deleted.

```
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$ echo "This is Divya Erupaka" > test
vboxuser@Ubuntu:~/Desktop$ cat test
This is Divya Erupaka
vboxuser@Ubuntu:~/Desktop$
```

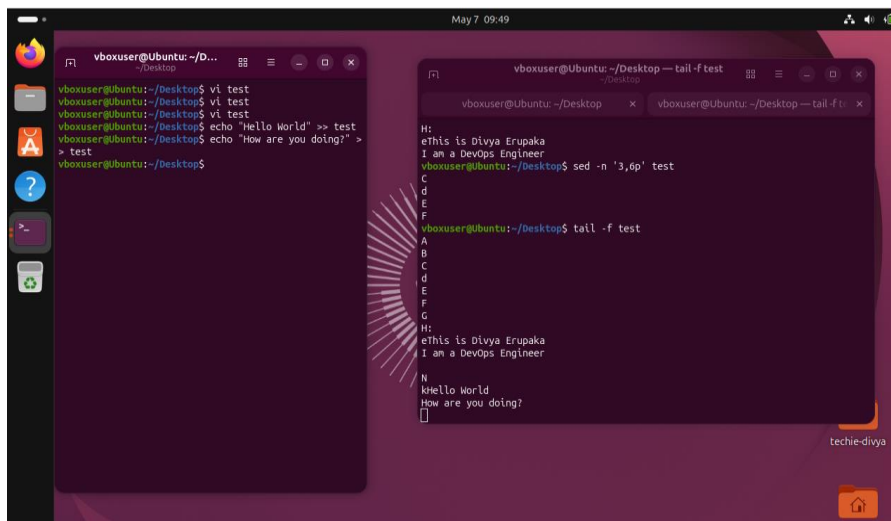
Double Redirection (>>) – I have run (echo "Hello Everyone" >> test) it will append the saved in the next line.

```
vboxuser@Ubuntu:~/Desktop$ echo "I am a DevOps Engineer" >> test
vboxuser@Ubuntu:~/Desktop$ cat test
This is Divya Erupaka
I am a DevOps Engineer
vboxuser@Ubuntu:~/Desktop$
```

If suppose we have 1000's of lines in a file and would just want to read first 3 lines , last 4 lines and the middle lines.

```
vboxuser@Ubuntu:~/Desktop$ vi test
vboxuser@Ubuntu:~/Desktop$ cat test
A
B
C
d
E
F
G
H:
eThis is Divya Erupaka
I am a DevOps Engineer
vboxuser@Ubuntu:~/Desktop$ head -3 test
A
B
C
vboxuser@Ubuntu:~/Desktop$ tail -4 test
G
H:
eThis is Divya Erupaka
I am a DevOps Engineer
vboxuser@Ubuntu:~/Desktop$ sed -n '3,6p' test
C
d
E
F
vboxuser@Ubuntu:~/Desktop$
```

Tail -f test command is used to monitor the file.

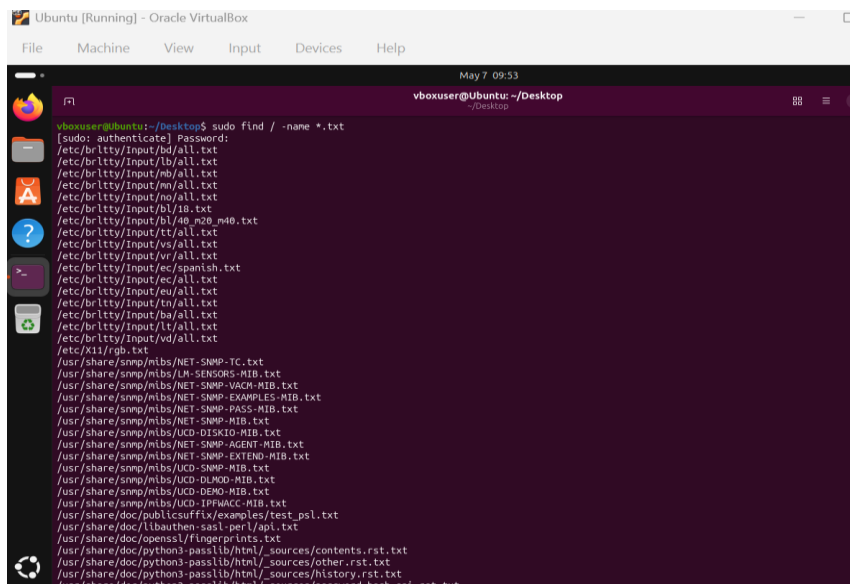


```
vboxuser@Ubuntu: ~/Desktop
vboxuser@Ubuntu:~/Desktop$ vi test
vboxuser@Ubuntu:~/Desktop$ vi test
vboxuser@Ubuntu:~/Desktop$ echo "Hello World" >> test
vboxuser@Ubuntu:~/Desktop$ echo "How are you doing?" >> test
vboxuser@Ubuntu:~/Desktop$

vboxuser@Ubuntu:~/Desktop$ tail -f test
H:
eThis is Divya Erupaka
I am a DevOps Engineer
vboxuser@Ubuntu:~/Desktop$ sed -n '3,6p' test
C
d
E
F
vboxuser@Ubuntu:~/Desktop$ tail -f test
A
B
C
d
E
F
C
H:
eThis is Divya Erupaka
I am a DevOps Engineer

N
kHello World
How are you doing?
[]
```

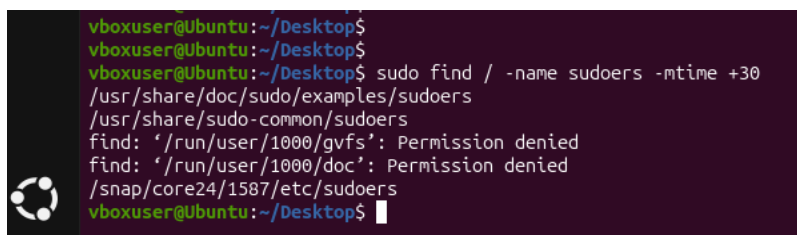
if not sure about the file name would like to know
sudo find / -name *.txt



```
Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

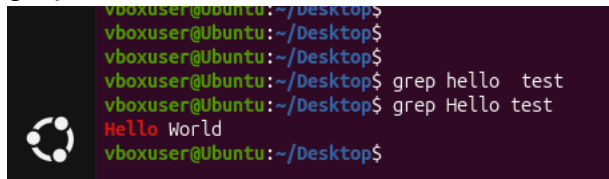
vboxuser@Ubuntu: ~/Desktop
vboxuser@Ubuntu:~/Desktop$ sudo find / -name *.txt
[sudo: authenticate] Password:
/etc/brlty/Input/bd/all.txt
/etc/brlty/Input/lb/all.txt
/etc/brlty/Input/nb/all.txt
/etc/brlty/Input/mn/all.txt
/etc/brlty/Input/no/all.txt
/etc/brlty/Input/bl/10.txt
/etc/brlty/Input/bl/40_m40.txt
/etc/brlty/Input/tc/all.txt
/etc/brlty/Input/vs/all.txt
/etc/brlty/Input/vr/all.txt
/etc/brlty/Input/ec/spanish.txt
/etc/brlty/Input/ec/all.txt
/etc/brlty/Input/eu/all.txt
/etc/brlty/Input/tn/all.txt
/etc/brlty/Input/be/all.txt
/etc/brlty/Input/lt/all.txt
/etc/brlty/Input/vd/all.txt
/etc/x11/rgb.txt
/usr/share/snp/mibs/NET-SNMP-TC.txt
/usr/share/snp/mibs/LM-SENSORS-MIB.txt
/usr/share/snp/mibs/NET-SNMP-VACM-MIB.txt
/usr/share/snp/mibs/NET-SNMP-EXAMPLES-MIB.txt
/usr/share/snp/mibs/NET-SNMP-PASS-MIB.txt
/usr/share/snp/mibs/NET-SNMP-MIB.txt
/usr/share/snp/mibs/UCD-DISKIO-MIB.txt
/usr/share/snp/mibs/NET-SNMP-AGENT-MIB.txt
/usr/share/snp/mibs/NET-SNMP-EXTEND-MIB.txt
/usr/share/snp/mibs/UCD-SNMP-MIB.txt
/usr/share/snp/mibs/UCD-DLMOD-MIB.txt
/usr/share/snp/mibs/UCD-DENO-MIB.txt
/usr/share/snp/mibs/UCD-IPPMACC-MIB.txt
/usr/share/doc/publicsuffix/examples/test_psl.txt
/usr/share/doc/libauthen-sasl-perl/apl.txt
/usr/share/doc/openssl/fingerprints.txt
/usr/share/doc/python3-passlib/html/_sources/contents.rst.txt
/usr/share/doc/python3-passlib/html/_sources/other.rst.txt
/usr/share/doc/python3-passlib/html/_sources/history.rst.txt
/usr/share/doc/python3-passlib/html/_sources/password_hash_api.rst.txt
```

To check the file older than 30 days
find / -name sudoers -mtime +30

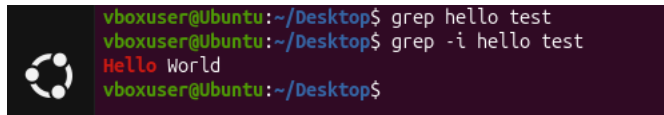


```
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$
vboxuser@Ubuntu:~/Desktop$ sudo find / -name sudoers -mtime +30
/usr/share/doc/sudo/examples/sudoers
/usr/share/sudo-common/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
/snap/core24/1587/etc/sudoers
vboxuser@Ubuntu:~/Desktop$
```

To Search for specific keyword from the file
grep Hello Test

A terminal window with a dark purple background and a white circular logo on the left. The text shows a user at the 'vboxuser@Ubuntu:~/Desktop\$' prompt. They enter 'grep hello test', which returns no output. Then they enter 'grep Hello test', which returns 'Hello World' in red text. The prompt returns to 'vboxuser@Ubuntu:~/Desktop\$'.

Because I used a small h, the grep command did not work due to case sensitivity.
To ignore uppercase and lowercase differences, use the grep -i option.

A terminal window with a dark purple background and a white circular logo on the left. The text shows a user at the 'vboxuser@Ubuntu:~/Desktop\$' prompt. They enter 'grep hello test', which returns no output. Then they enter 'grep -i hello test', which returns 'Hello World' in red text. The prompt returns to 'vboxuser@Ubuntu:~/Desktop\$'.